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4

Interfaces

C04: System & AI Essentials

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Join the lecture through the online portal

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Recap

What is this ?

What is this ?

a steering wheel, used to **control** and **guide** the direction of a vehicle.

We saw this!

SOURCE
1 2 3
4 5 6
7 8 9
TTX/MIX
0
PRE-CH
MUTE
VOL
CH LIST
CH
NETFLIX
video
Samsung TV Plus
GUIDE
RETURN
EXIT
A B C D
SETTINGS
INFO
BN59-01315M
SAMSUNG

We saw this!

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A TV remote, used to **control** and **interact** with a television from a distance.

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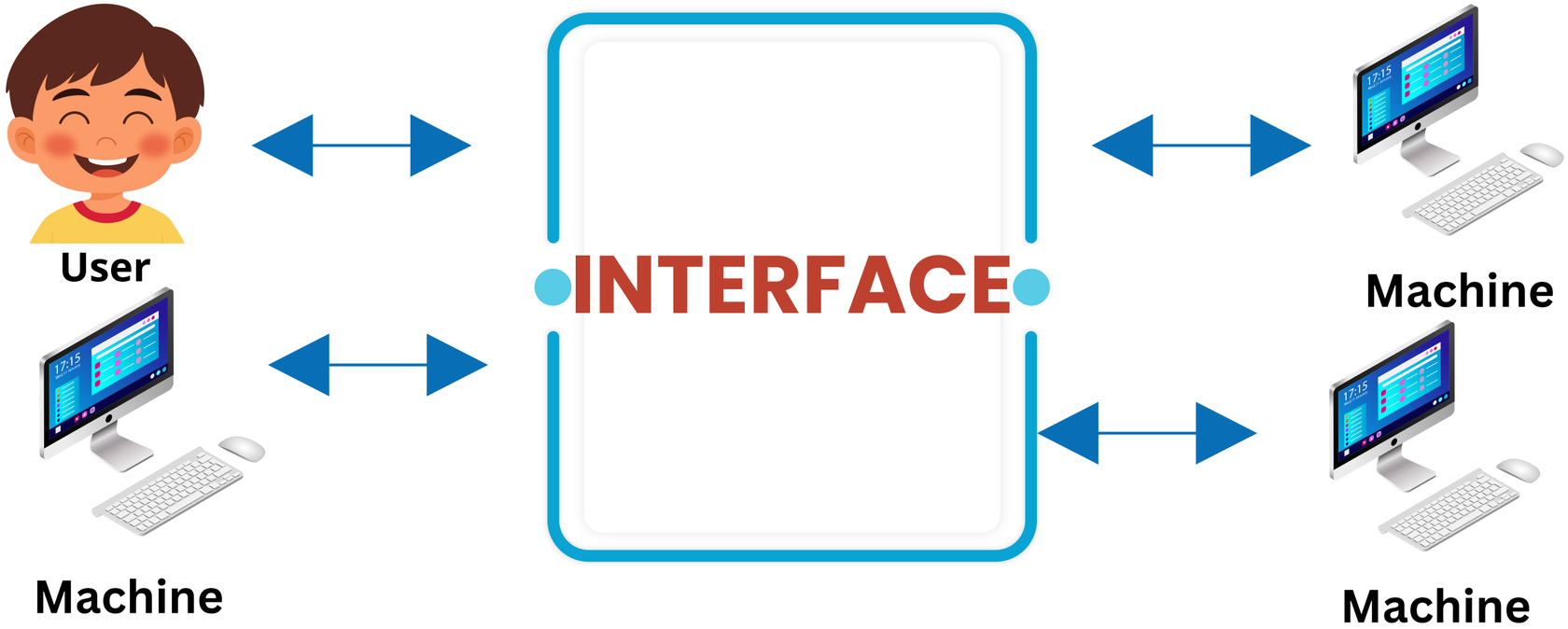
Both of these are **interfaces**.

A steering wheel is an interface between you and a car. A remote is an interface between you and a TV.

Interface

The layer between the **user** and the **machine**.

Think of an interface as a "face-to-face," a place where things, or people, or people and things (like you and your computer) meet. Any common boundary or area of convergence can be an interface.



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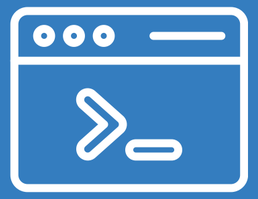
The Three Interfaces

The Three Interfaces



CLI

Command Line Interface

The Three Interfaces

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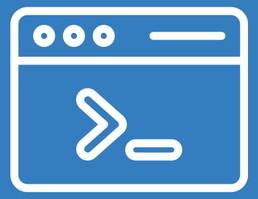

CLI

Command Line Interface

GUI

Graphical User Interface

The Three Interfaces

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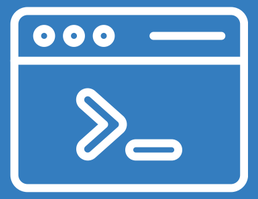


CLI

Command Line Interface

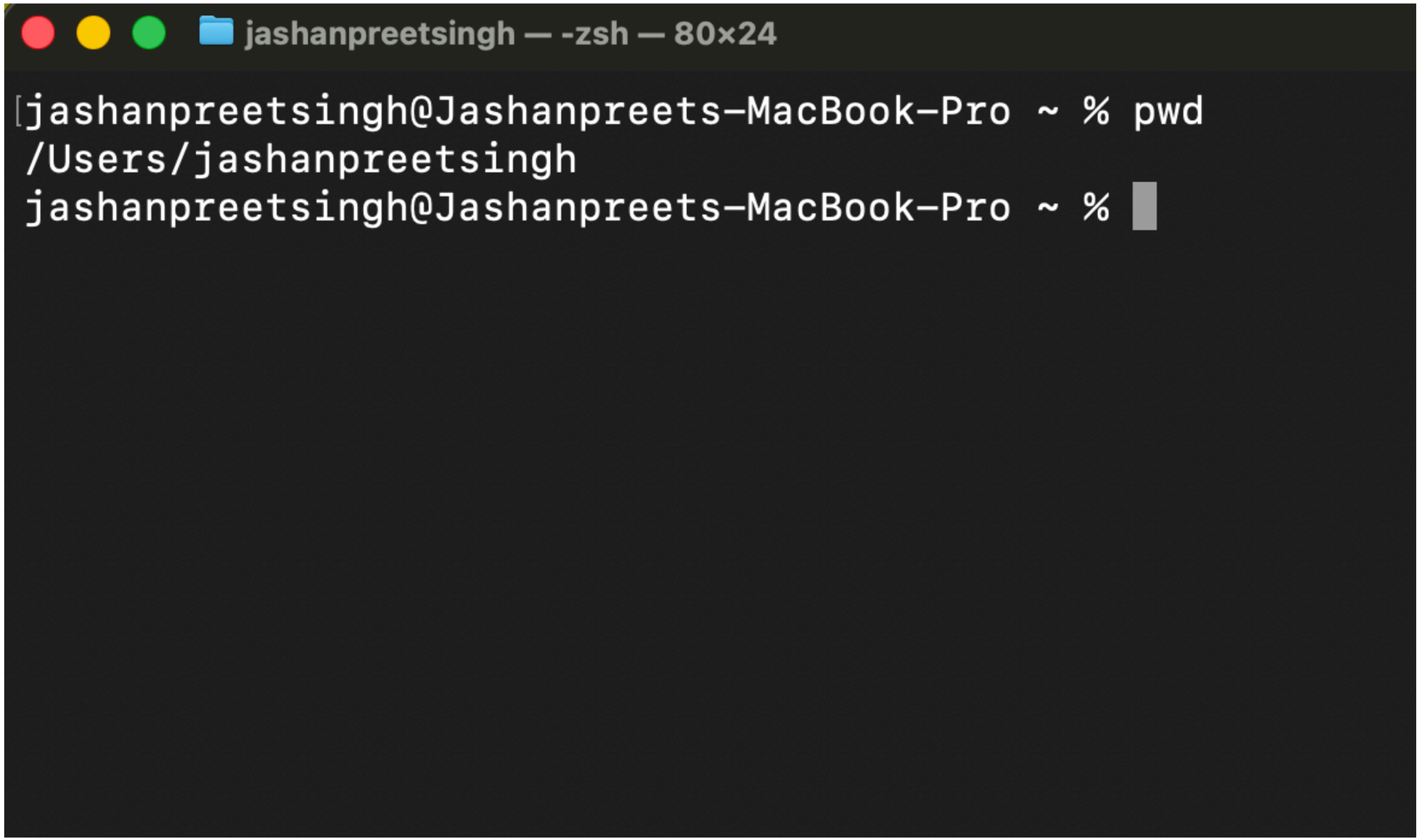
GUI

Graphical User Interface

API

Application Programming Interface

CLI - Command Line Interface



CLI – Command Line Interface

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Computers can use **text-based interaction.**

The OS provides a **terminal**: a prompt where you type commands and read responses. This is the CLI.

So, **CLI is a text-based** method for interacting with a computer's operating system or software.

Limitations of CLI

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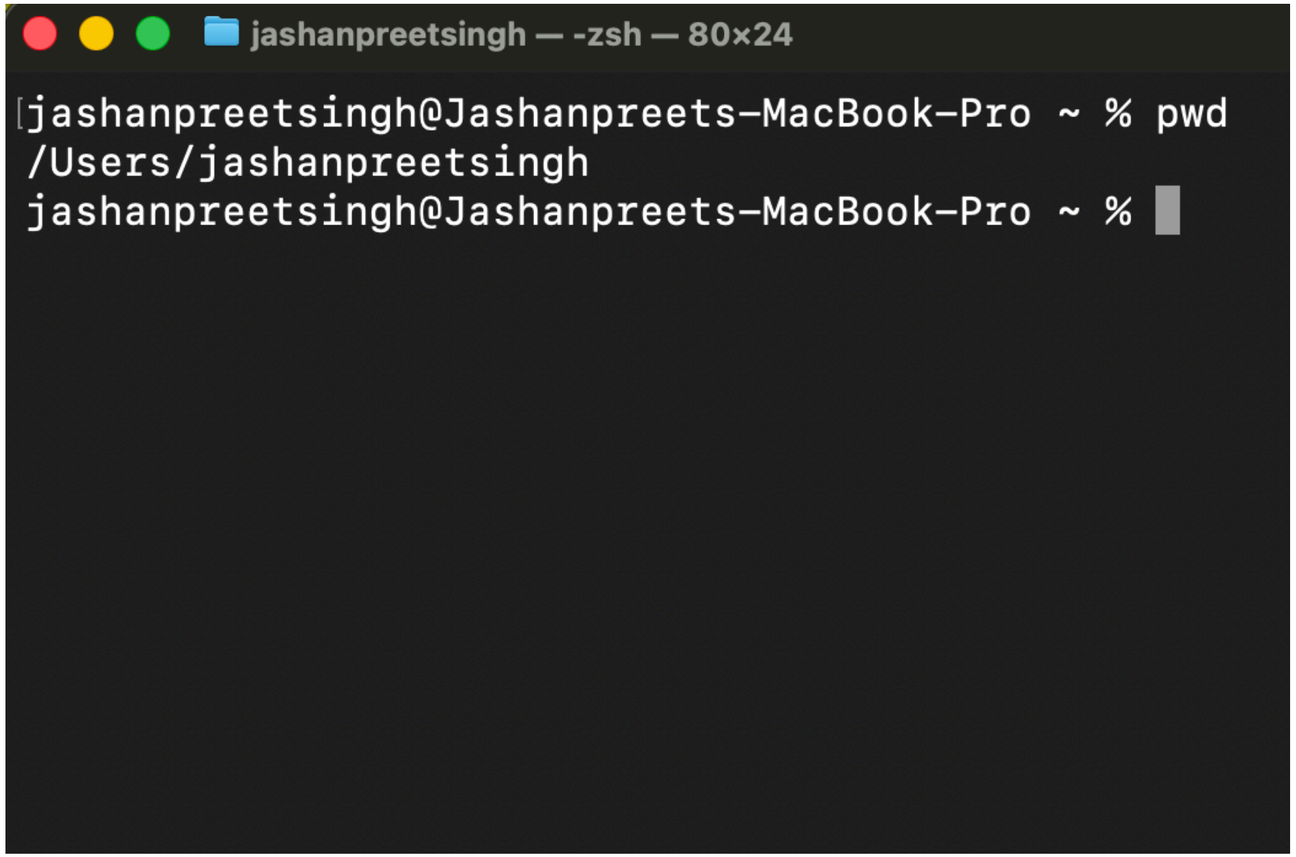
- Users must learn commands
- Difficult for beginners
- Small mistakes cause errors



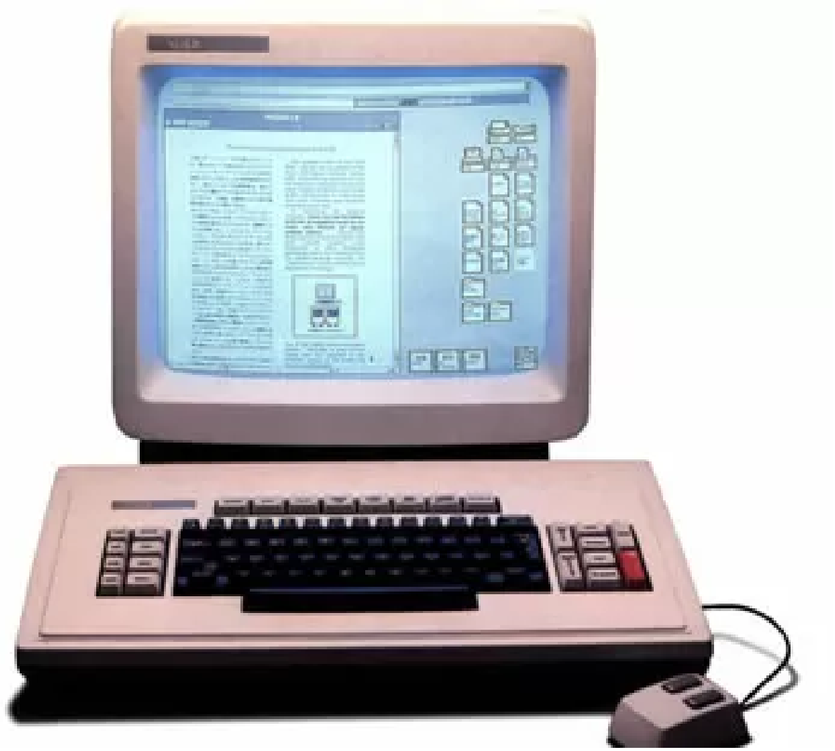
Think about it:

How would an average person use a computer if every action required memorizing **commands**?

htop for example

GUI – Graphical User Interface

A GUI (Graphical User Interface) allows users to **interact** with a computer through **visual elements** like windows, icons, buttons, and menus.



Birth of GUI:

The GUI emerged as a new way to interact with computers visually and intuitively.

- 1970s → **Xerox PARC**
- 1984 → **Macintosh**

GUI – Graphical User Interface

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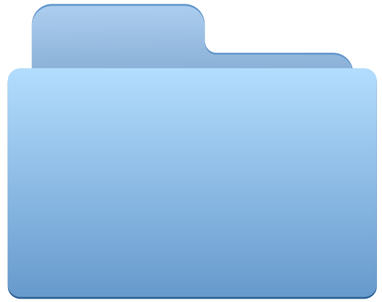
The computer was designed to feel like a real desk.

- **Files → Paper**: Files were designed to look like real paper documents.
- **Folders → Real Folders**: Folders visually organized files like physical office folders.
- **Delete → Trash Can**: Deleting a file was represented by moving it to a trash can.



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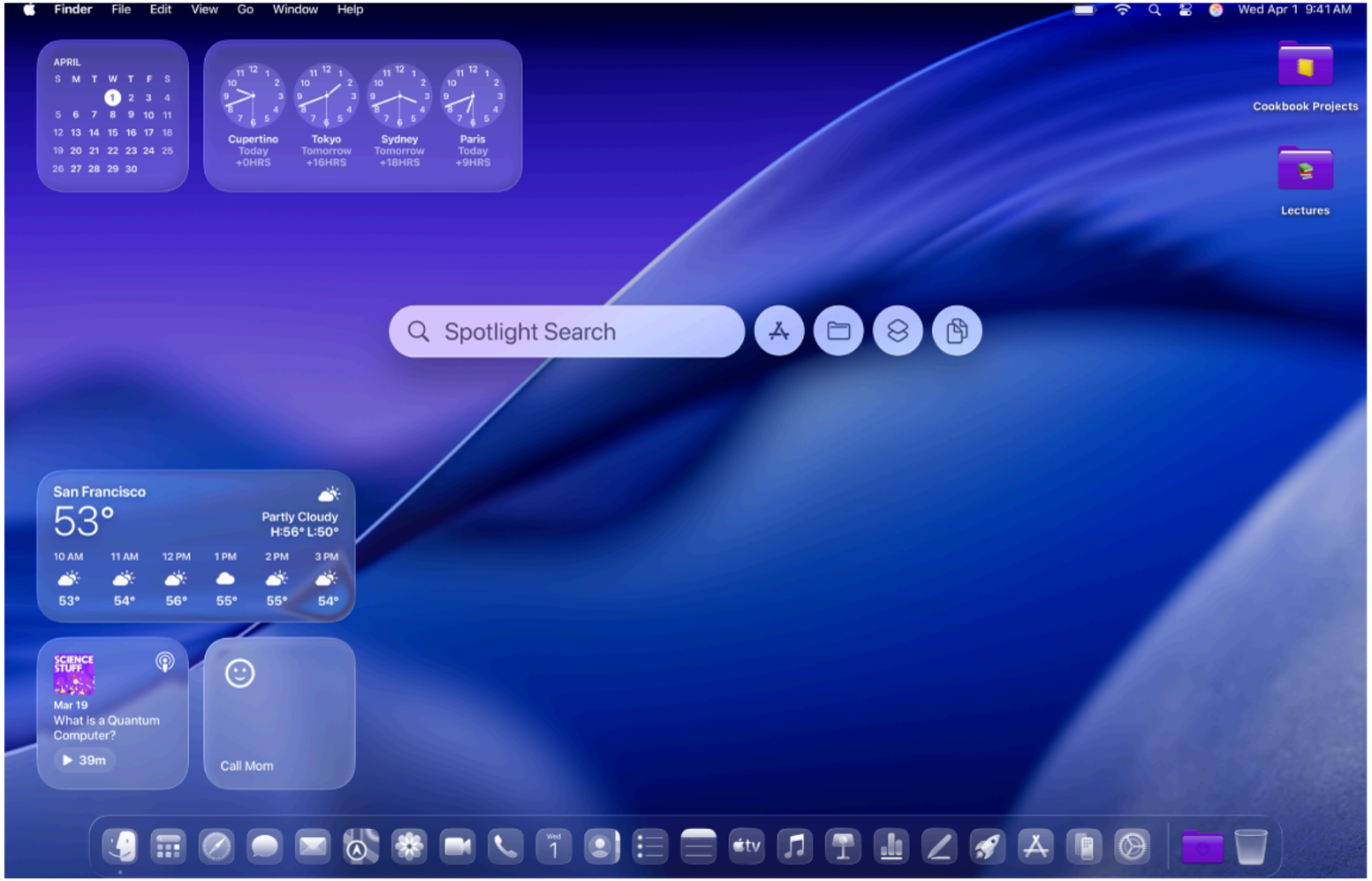
Let's explore what makes a good interface.

What Makes a GUI “Good”?



What Makes a GUI “Good”?

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- **Hides Complexity:** Shows only what the user needs right now.
- **Makes the Right Action Obvious:** Clear buttons, hierarchy, and navigation guide the user naturally.
- **Gives Feedback:** Every action responds visually loading, success, errors, hover states.
- **Speaks the User’s Language:** Uses familiar words, icons, and real-world metaphors instead of technical terms.

GUI design depends on humans decisions

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A GUI is only as good as the **decisions** the designer made. The OS gave them a canvas. What they painted was their **responsibility**.

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How do two applications communicate with each other ?

API – Application Programming Interface

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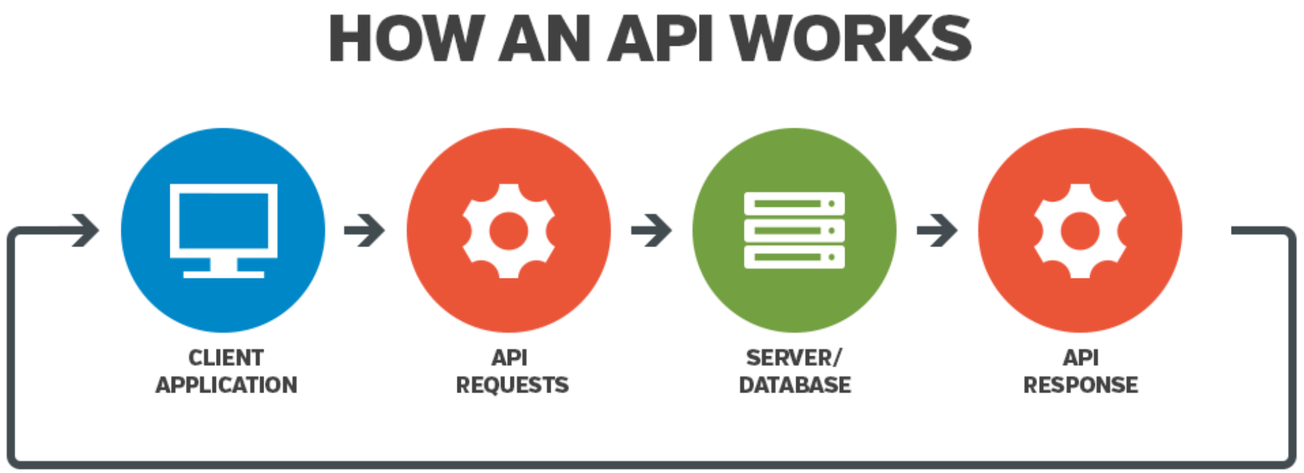
An API works like a **restaurant** menu. Customers **place requests** through the menu, and the kitchen **returns the result**. Similarly, software systems **communicate** through APIs without accessing each other's internal workings.

The menu is the interface between the customer and the kitchen.

What is an API ?

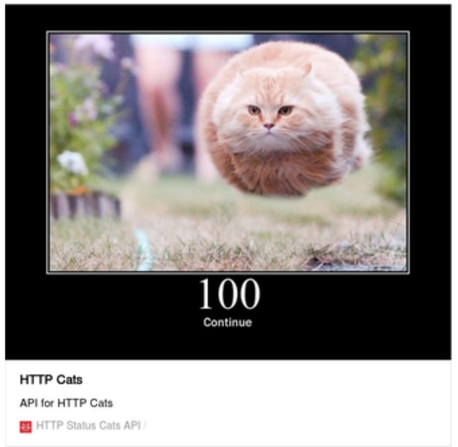
- APIs allow one program to request **services** from another.
- No buttons or graphical screens are required.
- One system sends a request and receives a response.

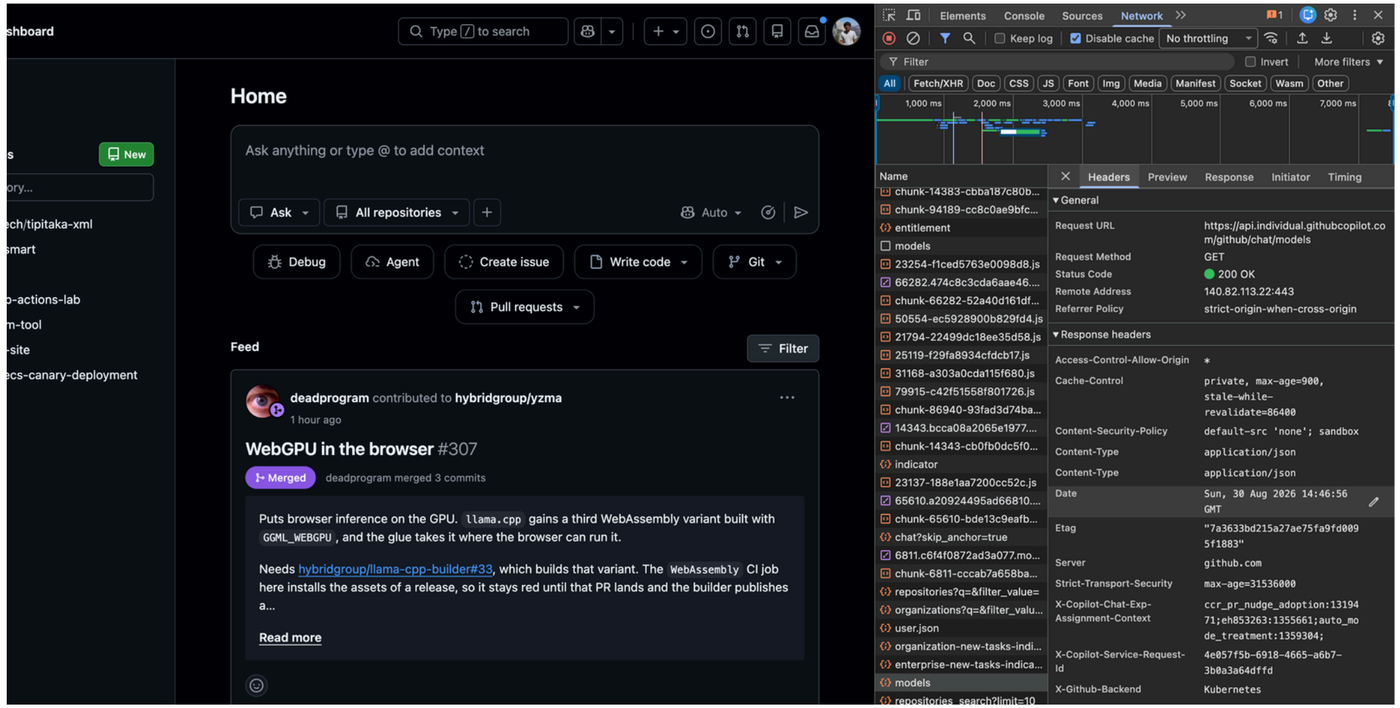


What is an API ?

- curl http://api.open-notify.org/iss-now.json

{"timestamp": 1788100039, "iss_position": {"longitude": "-128.3012", "latitude": "41.0496"}, "message": "success"}





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Inclass Quiz

Feedback

Now it's time to give your valuable feedback

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Thank You!

Further Reading

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- **How Two Programs Communicate: CLI and GUI Interactions** [https://blog.devgenius.io/how-two-programs-communicate-cli-and-gui-interactions-151844b6c845]
- **What Is a Command-Line Interface?** [https://www.coursera.org/articles/command-line-interface]
- **What Is a Graphical User Interface (GUI)?** [https://www.coursera.org/in/articles/gui]
- **What Is an API (Application Programming Interface)?** [https://www.ibm.com/think/topics/api]
- **Difference Between CLI and GUI** [https://www.geeksforgeeks.org/operating-systems/difference-between-cli-and-gui/]

Errors as Feedback

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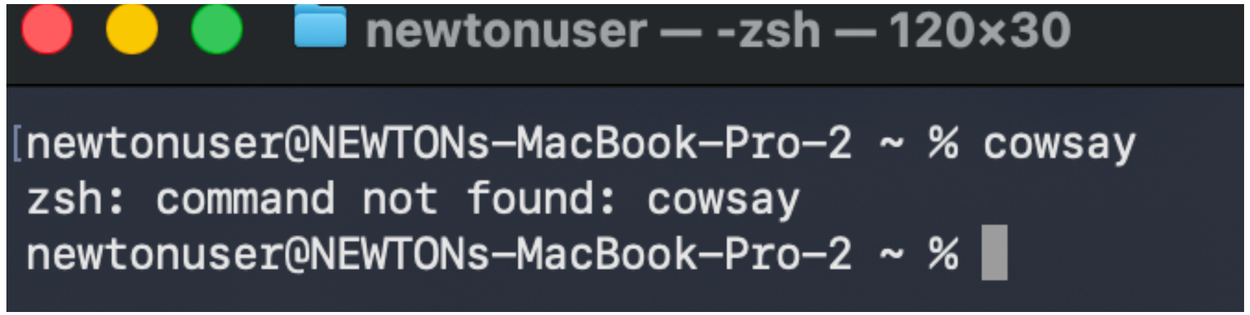
Every layer in the system can respond when something goes **wrong**.



Different Layers, Different Errors

- Shell → command errors
- Applications → crashes or pop-ups
- APIs → response/status errors
- Kernel → system-level failures

Each layer communicated problem differently.